

☐

I'm not robot

  
reCAPTCHA

Continue



# Introduction to probability bertsekas and tsitsiklis pdf

**Bertsekas and tsitsiklis introduction to probability pdf. Introduction to probability problems and solutions. Introduction to probability and statistics solutions. Introduction to probability bertsekas solutions. Introduction of probability and statistics.**

Page 1 Introduction to Probability SECOND EDITION Dimitri P. Bertsekas and John N. Tsitsiklis Massachusetts Institute of Technology WWW site for book information and orders – Athena Scientific, Belmont, Massachusetts Athena Scientific Post Office Box 805 Nashua, NH 03061-0805 U.S.A. Email: [email protected] WWW: Cover Design: Ann Gallagher © 2002, 2008 Dimitri P. Bertsekas and John N. Tsitsiklis All rights reserved. No part of this book may be reproduced in any form by any electronic or mechanical means (including photocopying, recording, or in forma tina storage and retrieval) without permission in writing from the publisher. Publisher's Cataloging-in-Publication Data Bertsekas, Dimitri P., Tsitsiklis, John N. Introduction to Probability Includes bibliographical references and index I. Probabilities. 2. Stochastic Processes. I. Title. QA273.B475 2008 519.2. – 21 Library of Congress Control Number: 2002092167 ISBN 978-1-86529-23-6 To the memory of Pantelis Bertsekas and Nikos Tsitsiklis Preface Probability is common sense reduced to calculation Laplace This book is an outgrowth of our involvement in teaching an introductory prob ability course ("Probabilistic Systems Analysis" ♦) at the Massachusetts Institute of Technology. The course is attended by a large number of students with diverse back grounds, and a broad range of interests. They span the entire spectrum from freshmen to beginning graduate students, and from the engineering school to the school of management. Accordingly, we have tried to strike a balance between simplicity in exposition and sophistication in analytical reasoning. Our key aim has been to develop the ability to construct and analyze probabilistic models in a manner that combines intuitive understanding and mathematical precision. In this spirit, some of the more mathematically rigorous analysis has been just sketched or intuitively explained in the text. so that complex proofs do not stand in the way of an otherwise simple exposition. At the same time, some of this analysis is developed (at the level of advanced calculus) in theoretical prob lems, that are included at the end of the corresponding chapter. Furthermore, some of the subtler mathematical issues are hinted at in footnotes addressed to the more attentive reader.

MA2040: Probability, Statistics and Stochastic Processes  
Problem Set-I

### Problem Set-

Sivaram Ambikavar

**1.** A six-sided die is made in a way that each even face is twice as likely as each odd face. All even faces are equally likely, as are all odd faces. Construct a probabilistic model for a single roll of this die and find the probability that the outcome is less than 4.

**Solution:** The sample space is  $\Omega = \{1, 2, 3, 4, 5, 6\}$ . Let the probability of each odd face be  $p$ . We then have the probability of each even face to be  $2p$ . We have

$$1 = P(\Omega) = P(\{1\}) + P(\{2\}) + P(\{3\}) + P(\{4\}) + P(\{5\}) + P(\{6\})$$

This gives us that

$$\mathbb{P}(\text{Outdegree is less than } k) = \mathbb{P}(i(1)) + \mathbb{P}(i(2)) + \mathbb{P}(i(3)) = p + 3p + p = 5p = 1/2$$

2. Let  $S_1, S_2, \dots, S_r$  be a partition of the sample space  $\Omega$ .

(a) Show that for any event  $A$ ,

$$P(A) = \sum_{i=1}^n P(A \cap S_i)$$

**Solution:** Since  $S_1, S_2, \dots, S_n$  is a partition of  $\Omega$ , we have  $\Omega = \bigcup_{i=1}^n S_i$  and  $S_i \cap S_j = \emptyset$ .

$$i \neq j. \text{ We have } \dots \otimes \dots \otimes (\tilde{u}_i \otimes \dots) \otimes (\tilde{u}_j \otimes \dots)$$
$$A = A|_M \cap A|_N \cap \left( \bigcup_{i=1}^n \mathcal{R}_i \right) = \bigcup_{i=1}^n (A|_M \cap \mathcal{R}_i)$$

Since  $S_i$ 's are mutually disjoint, so are  $A \cap S_i$ 's. Hence, we have that

$$P(A) = P\left(\bigcup_{i=1}^{\infty} (A \cap S_i)\right) = \sum_{i=1}^{\infty} P(A \cap S_i)$$
$$r(\mathcal{A}) = \sum_{i=1}^n \left( \sum_{j=1}^n a_{ij} \right) = \sum_{j=1}^n \left( \sum_{i=1}^n a_{ij} \right)$$

(b) Use the previous part to show that, for events  $A$ ,  $B$  and  $C$ ,

$$P(A) = P(A \cap B) + P(A \cap C) + P(A \cap B \cap C) - P(A \cap B \cap C)$$

**Solution:** Note that our sample space can be partitioned into two disjoint sets  $S_1 = \{B\}$  and  $S_2 = \{B^c\} = B \cap C^c$ . From intuition, test, we hence have

$$\mathcal{H}(A) = \mathcal{H}(A \otimes B) = \mathcal{H}(A \otimes B) = \mathcal{H}(A \otimes (B \oplus C)) = \mathcal{H}(A \otimes (B \oplus C))$$
$$= \mathbb{P}(\{(A \cap B) \cup (A \cap C)\}) + \mathbb{P}(A \cap B^c \cap C^c)$$
$$= P(A \cap B) + P(A \cap C) - P((A \cap B) \cap (A \cap C)) + P(A \cap B^c \cap C^c)$$
$$= P(A \cap B) + P(A \cap C) + P(A \cap B^c \cap C^c) - P(A \cap B \cap C)$$

The book covers the fundamentals of probability theory (probabilistic models, discrete and continuous random variables, multiple random variables, and limit theorems), which are typically part of a first course on the subject. It also contains, in Chapters 4-6 a number of more advanced topics, from which an instructor can choose to match the goals of a particular course. In particular, in Chapter 4, we develop transforms, a more advanced view of conditioning, sums of random variables, least squares estimation, and the bivariate normal distribution. Furthermore, in Chapters 5 and 6, we provide a fairly detailed introduction to Bernoulli, Poisson, and Markov processes. Our M.L.T. course covers all seven chapters in a single semester, with the exception of the material on the bivariate normal (Section 4.7), and on continuous time Markov chains (Section 6.5). However, in an alternative course, the material on stochastic processes could be omitted, thereby allowing additional emphasis on foundational material, or coverage of other topics of the instructor's choice. Our most notable omission in coverage is an introduction to statistics. While we develop all the basic elements of Bayesian statistics, in the form of Bayes' rule for discrete and continuous models, and least squares estimation, we do not enter the subjects of parameter estimation, or non-Bayesian hypothesis testing. The problems that supplement the main text are divided in three categories: (a) Theoretical problems: The theoretical problems (marked by \*) constitute an important component of the text, and ensure that the mathematically oriented reader will find here a smooth development without major gaps. Their solutions are given in the text, but an ambitious reader may be able to solve many of them, especially in earlier chapters, before looking at the solutions.

(D) Problems in the text: Besides theoretical problems, the text contains several problems, of various levels of difficulty. These are representative of the problems that are usually covered in recitation and tutorial sessions at M.I.T., and are a primary mechanism through which many of our students learn the material. Our hope is that students elsewhere will attempt to solve these problems, and then refer to their solutions to calibrate and enhance their understanding of the material. The solutions are posted on the book's [www site](#) (c) Supplementary problems: There is a large (and growing) collection of additional problems, which is not included in the book, but is made available at the book's [www site](#). Many of these problems have been assigned as homework or exam problems at M.I.T., and we expect that instructors elsewhere will use them for a similar purpose. While the statements of these additional problems are publicly accessible, the solutions are made available from the authors only to course instructors. We would like to acknowledge our debt to several people who contributed in various ways to the book. Our writing project began when we assumed responsibility for a popular probability class at M.I.T. that our colleague Al Drake had taught for several decades. We were thus fortunate to start with an organization of the subject that had stood the test of time, a lively presentation of the various topics in AI's classic textbook, and a rich set of material that had been used in recitation sessions and for homework. We are thus indebted to Al Drake Preface vii for providing a very favorable set of initial conditions. We are thankful to the several colleagues who have either taught from the draft of the book at various universities or have read it, and have provided us with valuable feedback.

In particular, we thank Ibrahim Abou Faycal, Gustavo de Veciana, Eugene Feinberg, Bob Gray, Muriel Medard, Jason Papastavrou, Ilya Pollak, David Tse, and Terry Wagner. The teaching assistants for the M.I.T. class have been very helpful. They pointed out corrections to various drafts, they developed problems and solutions suitable for the class, and through their direct interaction with the student body, they provided a robust mechanism for calibrating the level of the material. Reaching thousands of bright students at M.J. T. at an early stage in their studies was a great source of satisfaction for us. We thank them for their valuable feedback and for being patient while they were taught from a textbook-in-progress. Last but not least, we are grateful to our families for their support through out the course of this long project. Dimitri P. Bertsekas, [email protected] John N. Tsitsiklis, [email protected] Cambridge, Mass., May 2002 v iii Preface Preface to the Second Edition This is a substantial revision of the 1st edition, involving a reorganization of old material and the addition of new material. The length of the book has increased by about 25 percent. The main changes are the following: (a) Two new chapters on statistical inference have been added, one on Bayesian and one on classical methods. Our philosophy has been to focus on the main concepts and to facilitate understanding of the main methodologies through some key examples. (b) Chapters 3 and 4 have been revised, in part to accommodate the new material of the inference chapters and in part to streamline the presentia tion. Section 4.7 of the 1st edition (bivariate normal distribution) has been omitted from the new edition, but is available at the book's website. (c) A number of new examples and end-of-chapter problems have been added. The main objective of the new edition is to provide flexibility to instructors in their choice of material, and

in particular to give them the option of including an introduction to statistical inference. Note that Chapters 6-7, and Chapters 8-9 are mutually independent, thus allowing for different paths through the book. Furthermore, Chapter 4 is not needed for Chapters 5-7, and only Sections 4.2-4.3 from Chapter 4 are needed for Chapters 8 and 9. Thus, one possible course offerings based on this book are: (a) Probability and introduction to statistical inference: Chapters 1-3, Sections 4.2-4.3, Chapter 5, Chapters 8-9. (b) Probability and introduction to stochastic processes: Chapters 1-3 and 5-7, with possibly a few sections from Chapter 4. We would like to express our thanks to various colleagues who have contributed valuable comments on the material in the 1st edition and/or the reorganization of the material in the new chapters. Ed Coffman, Munther Dahleh, Vivek Goyal, Anant Sahai, David Tse, George Verghese, Alan Willsky, and John Wyatt have been very helpful in this regard. Finally, we thank Mengdi Wang for her help with figures and problems for the new chapters. Dimitri P. Bertsekas, [email protected] John N. Tsitsiklis, [email protected] Cambridge, Mass., June 2008 Contents 1. Sample Space and Probability 1.1. Sets . . .

# Introduction to Probability

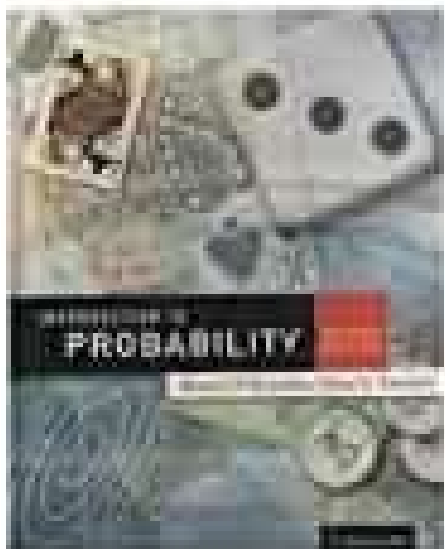
SECOND EDITION

Dimitri P. Bertsekas and John N. Tsitsiklis

WWW site for book information and  
<http://www.athenasc.com>

Athena Scientific, Belmont, Massachusetts

..... 1.2. Probabilistic Models ..... 1.3. Conditional Probability ...  
1.4. Total Probability Theorem and Bayes' Rule 1.5. Independence ....



1.6. Counting . . . . . 1.7. Summary and Discussion Problems . . . . . 2. Discrete Random Variables 2.1. Basic Concepts . . . . . 2.2. Probability Mass Functions 2.3. Functions of Random Variables 2.4. Expectation, Mean, and Variance 2.5. Joint PMFs of Multiple Random Variables 2.6. Conditioning . . . . . 2.7. Independence . . . . . 2.8. Summary and Discussion Problems . . . . .

3. General Random Variables 3.1. Continuous Random Variables and PDFs 3.2. Cumulative Distribution Functions 3.3. Normal Random Variables . . . . . 3.4. Joint PDFs of Multiple Random Variables 3.5. Conditioning . . . . . 3.6. The Continuous Bayes' Rule . . . . . p. 1 . . . . . p. 3 . . . . . p. 6 p. 18 p. 28 p.

